ITAI-1371 - Intro to Machine Learning

Reflective Journal

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Most of the code for Lab 07 was already provided in the Google Colab notebook, so I did not build the entire machine learning workflow from scratch. My hands-on contribution was to fix the line breaks in the setup and model training code, update the line that fills missing ages with the median so that the result was assigned directly to the Age column, and run the code. I also added print(y_pred[:10]) to display the first ten predictions. Seeing zeros and ones helped me confirm that the model was producing the expected type of output for a binary classification task.

This verification also helped me understand the difference between a program running and a model performing well. The first ten predictions showed that the code generated class labels, but they did not tell me whether the predictions were accurate. The notebook's evaluation outputs were needed to examine model performance. For example, the confusion matrix showed both correct predictions and mistakes. The classification report gave additional information through precision, recall, and F1-score.

I learned that precision and recall help describe different kinds of classification errors. Precision matters when false positive predictions are costly, while recall matters when missing positive cases is more serious. The Knowledge Check examples made this easier to understand: avoiding important emails being incorrectly marked as spam is a precision concern, while detecting as many serious illnesses as possible is a recall concern.

The cross-validation section also showed why a single train-test split may not give the full picture. The model's accuracy on one test split was 80%, while its average cross-validation score was about 78.68%. Comparing results from several folds gives a broader estimate of how consistently the model performs. One question I still have is how the evaluation results would change if we selected different features or handled missing values another way.

This lab helped me follow how the steps in a prepared notebook connect: clean and prepare the data, train the model, generate predictions, and evaluate those predictions. My contribution focused on correcting and checking parts of that process, while the provided outputs helped me understand the later evaluation steps.

Danh Nguyen

For me, the biggest realization was that our 78% accuracy hid a 0.64 recall for survivors. This reminded me of security alert systems, where a detector can look accurate overall but still miss many real attacks.

Great Inioluwa Olayiwola

Navigating Workflow

Complexity and Initial Impressions The syntax and libraries in machine learning appeared daunting. It felt as though I was looking at different stages of a puzzle with no clear picture to fit the pieces into. Yet, I found it interesting that the work done in data preparation, model training, and evaluation, formed a cohesive process. I could see the need and rationale for every step from data preparation and split to model performance evaluation. Now, I see machine learning as a series of structured and linked activities and not as disconnected lines of coding.

Breakthroughs in Concept Understanding and Evaluation

For the first time, confusion matrices and classification reports helped me move beyond the metrics that define accuracy. Seeing how my models behaved on actual cases with false negatives and true positives put the abstract concepts of statistical modeling in the context of the real world. Equally important was the transition from the single train-test split to a 5-fold crossvalidation. For the first time, I could clearly see the trade-offs of using one data split. Crossvalidation demonstrated how data could be lucky or unlucky, providing an essential level of reliability and robustness to my models.

Challenges Faced and Problem-Solving Strategies

Dealing with fragmented snippets and undefined variables is extremely difficult. My solution was a methodical, disciplined process. I had to rethink the data pipeline, confirm all variable assignments (especially cm and report), and cross reference the input expected by all plotting functions. As I corrected my data pipeline, I recognized the importance of re-executing all data processing cells to avoid frustrating and easily fixable syntax errors

Shifts in Thinking and Future Outlook

Programming changed for me after learning about algorithmic programming. Machine learning models aren't black boxes for me. They depend on careful data structuring, thoughtful feature selection, and precise metric evaluation. Moving forward, I will look at the way data is structured and examine how metrics affect real world decision-making.

Karima Sadruddin

Quan Thach

Working on this module shifted the analytical focus from baseline model building to exploring robust evaluation strategies like confusion matrices, classification reports, and cross-validation techniques. Progressing through the code configuration demonstrated that measuring a classification model solely through global accuracy provides an incomplete picture of its actual predictive utility. Writing the implementation for the confusion matrix explicitly highlighted where errors were clustered, separating true performance into isolated quadrants of false alarms and missed instances.

Interpreting the individual components of precision and recall reinforced how distinct business and engineering constraints demand completely different optimization criteria. Investigating metrics through the classification report made it clear that a model can perform excellently at identifying one specific class while failing to capture the nuances of another. This segment of the lab illustrated why balancing true positives against type one and type two errors determines whether a predictive workflow can be safely deployed in real-world environments.

Finally, executing the cross-validation assignment highlighted how much metrics can fluctuate depending on data partitioning choices. Watching the individual scores vary across separate folds proved that relying on a singular train-test partition presents a major risk of generating misleadingly optimistic performance signals. Aggregating those separate iterations into an average score with a measured standard deviation established a far more transparent, dependable strategy for confirming true generalization capability.